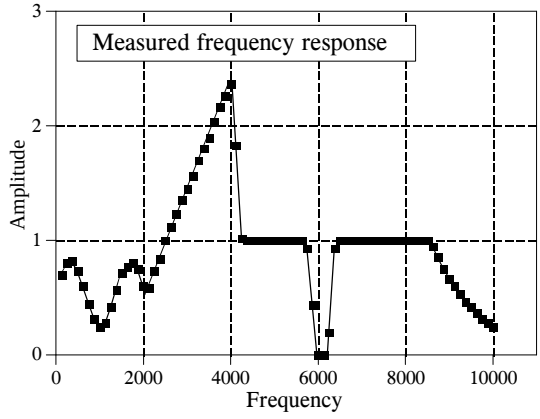

FIGURE 29-4
Measured frequency response. This graph shows measured points on the frequency response of the example FIR filter. These measured points have far less accuracy than the designed frequency response of Fig. 29-3a.

Our analog engineer will become very agitated when he sees this linear phase. The signals won't appear the way he thinks they should, and he will start twisting knobs at lightning speed. He will complain that the triggering isn't working right, and mumble such things as: "this doesn't make sense," what's going on here?", and "who's been playing with my oscilloscope?" The performance of DSP systems is so good, it will take him a few minutes before he understands what he is seeing.

To make him even more impressed, we ask our engineer to manually measure the frequency response of the system. To do this, he will step the signal generator through all the frequencies between 125 Hz and 10 kHz in increments of 125 Hz. At each frequency he measures the amplitude of the output signal and divides it by the amplitude of the input signal. (Of course, the easiest way to do this is to keep the input signal at a constant amplitude). We set the sampling rate of the EZ-KIT Lite at 22 kHz for this test. In other words, the 0 to 0.5 digital frequency of Fig. 29-2a is mapped to DC to 11 kHz in our real world measurement.

Figure 29-4 shows actual measurements taken on the EZ-KIT Lite; it couldn't be better! The measured data points agree with the theoretical curve within the limit of measurement error. This is something our analog engineer has *never* seen with filters made from resistors, capacitors, and inductors.

However, even this doesn't give the DSP the credit it deserves. Analog measurements using oscilloscopes and digital-volt-meters have a typical accuracy and precision of about 0.1% to 1%. In comparison, this DSP system is limited only by the ~0.001% round-off error of the 16 bit codec, since the internal calculations use floating point. In other words, the device being evaluated is *one-hundred times* more precise than the measurement tool being used. A proper evaluation of the frequency response would require a specialized instrument, such as a computerized data acquisition system with a 20 bit ADC. Given these facts, it is not surprising that DSPs are often used in measurement instruments to achieve high precision.